Class Imbalance in Histopathology: Which Classes Share the Tail Moves the Damage Threefold on TCGA-UT

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Abstract

At an imbalance ratio of $\rho = 100$ on TCGA-UT, a random assignment of the 30 cancer types to the ranks of a long-tailed training allocation costs 2.76 points of test balanced accuracy. Earlier studies of this series traced the dependence of that damage on the assignment to pairs of classes the balanced classifier confuses, but a first attempt to turn this into a score over class orders failed its own pre-registered check. This report repairs that score with one change: each confused pair is weighted by the absolute rate at which the balanced classifier sends one class to the other, instead of by that pair's share of the class's errors. The repaired score passes the same check on damages already observed, and a calibration line fitted on 30 random orders predicts both sorted orders of the companion tail-assignment study inside their confidence intervals. Three class orders derived from the score were then fitted, 30 fits each. The order that maximizes the score costs 4.17 points, 1.41 points (95 percent interval 0.94 to 1.88) more than a random order; the order that minimizes it costs 1.28 points, 1.48 points (1.04 to 1.92) less. Reversing the worst order, which keeps every confused pair exactly as far apart in rank but swaps which member sits at the smaller allocation, costs 1.09 points (0.42 to 1.78) less than the worst order, so the direction of separation matters and not only its size. The calibration line, fixed before these fits, predicts all three within 0.30 points. At a fixed ratio and a fixed budget, the choice of which classes are rare therefore moves the damage over a range of 2.9 points, larger than the average damage of the ratio itself.

1 Motivation

The TCGA-UT prevalence study measured how balanced accuracy on 30 cancer types degrades as the training class distribution moves from balanced to long-tailed, at twenty patients per class and 19,200 training patches held fixed (Queisler, 2026b). At $\rho = 100$ the damage was 2.76 points, averaged over class orders drawn at random per fit. The tail-assignment study then fixed two orders, sorting the classes by their recall under balanced training and placing either the easiest or the hardest classes in the tail (Queisler, 2026d). Both cost less than a random order, 2.35 and 1.56 points, and no class collapsed. What moved a class's recall was its allocation relative to the classes it is confused with: rectum adenocarcinoma lost 13.5 points of recall while sitting just below colon adenocarcinoma. That study named the natural worst case, an assignment that separates strongly confused pairs and puts the member with more recall to lose underneath, but did not fit it.

The worst-assignment study wrote that candidate as a directed pair-separation score and checked the score against the three damages already known before spending any fit on it (Queisler, 2026e). The score failed. It ranked the easy-tail order above every random order, although the easy-tail order is milder than the random average. The report traced the failure to the weights. Each class's confusion weights were normalized to sum to one, so a class that the balanced classifier almost never gets wrong carried as much total weight as a class it confuses

constantly. Sorting the classes by recall then aligned the remaining weight, recall itself, with rank, and the score rewarded that alignment rather than the separation of confused pairs.

This report keeps the score and the search of the worst-assignment study and changes only the weight: a pair now enters with the absolute rate at which the balanced classifier predicts one class for the other. It asks whether the order that maximizes the repaired score is more damaging than a random order, whether the order that minimizes it is milder, and whether the direction of separation, not only its size, contributes. Section 2 states the score, the orders it produces, and the arms fitted; Section 3 states the gate and the pre-registered endpoints; the results follow in Section 4.

2 Design

The study reuses the design of the TCGA-UT prevalence curve (Queisler, 2026b): 30 cancer types with frozen Virchow2 features (Zimmermann et al., 2024), three patient-disjoint splits with ten draws each, $G = 20$ patients per class, a total budget of $T = 19{,}200$ training patches, and nested patch orderings within each patient. The balanced arm r1 gives each class 640 patches. The ratio arm rρ gives the class at rank j a count proportional to $\rho^{-j/29}$, so at $\rho = 100$ the counts fall from 2,843 patches at rank 0 to 28 patches at rank 29. Only the assignment of classes to ranks varies between arms; patients, patch orderings, and per-rank counts are held fixed, so any difference in damage between two arms is due to the assignment alone. The classifier is multinomial logistic regression with regularization selected on validation patient-macro balanced accuracy (Queisler, 2026a).

2.1 A rate-weighted pair-separation score

Each rank r holds n_r training patches, and the score measures allocations on a logarithmic scale relative to the balanced arm, $z_r = \log_2(n_r/640)$. The quantity z_r is not a patch count. It counts how many times the allocation at rank r doubles or halves the balanced 640 patches, so it is positive for ranks that hold more than 640 patches and negative for ranks that hold fewer. At $\rho = 100$, rank 0 holds 2,843 patches, $z_0 = \log_2(2843/640) = 2.15$, and rank 29 holds 28 patches, $z_{29} = \log_2(28/640) = -4.51$. A difference $z_{r_d} - z_{r_c}$ is therefore the number of doublings by which the allocation of class d exceeds that of class c. An assignment π places class c at rank r_c. The score is

$$S(\pi) = \sum_{c \neq d} \underbrace{w_{c \to d}}_{\substack{\text{how often } c \\ \text{is predicted as } d}} \underbrace{h_c}_{\substack{\text{recall } c \\ \text{can lose}}} \underbrace{\max(z_{r_d} - z_{r_c},\, 0)}_{\substack{\text{doublings by which } d\text{'s} \\ \text{allocation exceeds } c\text{'s}}}, \qquad w_{c \to d} = \frac{N_{cd}}{\sum_e N_{ce}}, \tag{1}$$

where N_{cd} counts the test patches of class c that the balanced arm predicts as class d, pooled over all fits, and h_c is class c's pooled balanced-arm recall in percent. The weight $w_{c \to d}$ is thus the probability that the balanced classifier predicts d for a patch of class c. Each term measures the exposure of class c to a class d it is mistaken for that holds a larger allocation, scaled by how much recall c has to lose.

Figure 1 illustrates single terms with rectum adenocarcinoma (READ) as class c. READ is often predicted as colon adenocarcinoma (COAD). With READ at rank 27 and COAD at rank 2, COAD holds 5.73 doublings more patches, and the term is large (panel a). The term is small when the two classes sit at adjacent ranks (panel b). It is zero when READ sits above COAD, because the max keeps only the direction in which the confusion partner holds more patches (panel c); the reverse term, COAD predicted as READ, then becomes active instead, with its own weight and recall. The term is also near zero when READ sits far below a class it is almost never predicted as, such as thyroid carcinoma (THCA), because $w_{c \to d}$ is then close

to zero (panel d). A high-scoring pair is thus a pair that the balanced classifier confuses often, placed far apart, with the more often confused member at the smaller allocation.

The score of a whole assignment adds these terms over all 870 ordered pairs of classes. An assignment scores high when many strongly confused pairs are separated at the same time, each with the right direction; one assignment can satisfy many pairs at once because each pair constrains only the relative ranks of its two members. The bottom panels of Figure 1 show the result for the two extreme orders of Section 2.2. In the worst order, nearly every class sits far below the class it is most often predicted as. In the mild order, most confused classes sit next to their partner, and the remaining gaps are short.

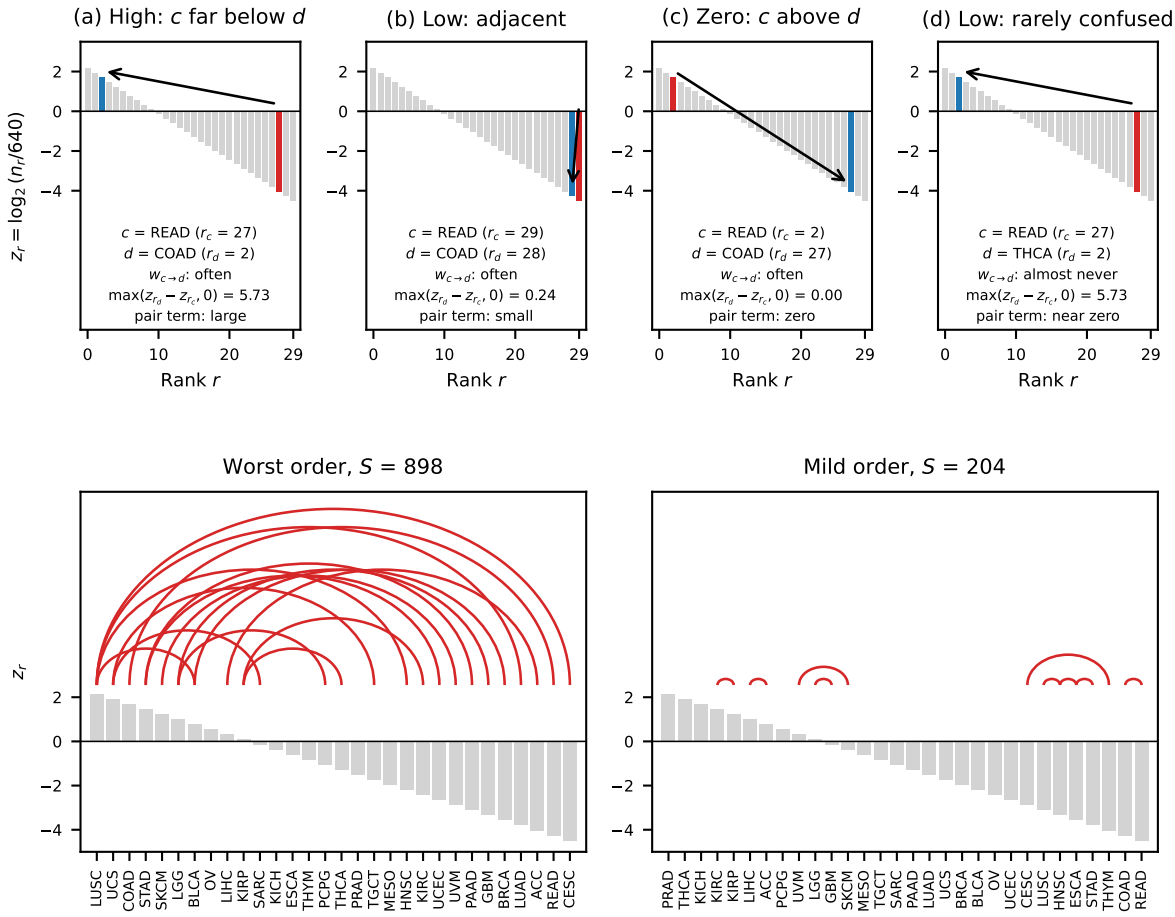


Figure 1: A pair term is large only when a frequently confused class sits far below its confusion partner, and the worst order stacks many such terms. Grey bars: log allocation z_r of each rank at $\rho = 100$; positive z_r means more than the balanced 640 patches, negative z_r fewer. Top, (a) to (d) from left to right: one term $w_{c\to d}\, h_c \max(z_{r_d} - z_{r_c}, 0)$ of Equation (1), with class c in red and class d in blue; the arrow points from c to d. Bottom: the worst and mild orders, with class codes under their ranks. Each arc links a class to the class it is most often predicted as in the balanced arm and is drawn only when the class sits below that partner, that is, when its term is positive. Arcs show only the main partner of each class, whereas S sums over all pairs.

The formula is that of the worst-assignment study; only the denominator of w differs (Queisler, 2026e). There it was the off-diagonal row sum $\sum_{e\neq c} N_{ce}$, so every class carried total weight one. With the full row sum in the denominator, a class's weights sum to its patch-level error rate, and a class that is rarely confused contributes little no matter where it sits. The score remains directed: because the per-rank counts are log-linear in rank, reversing an order

preserves every pairwise rank gap, and only the one-sided max and the asymmetric weights $w_{c\to d}h_c$ distinguish an order from its reversal.

2.2 Derived orders and fitted arms

The score is maximized and minimized over the 30! assignments by pairwise-swap hill climbing from 50 seeded random starts, taking the best improving swap until no swap improves. The result is an extreme assignment found by local search, not a proven optimum. Three orders follow. The *worst* order maximizes S. The *mild* order minimizes it. The *flip* order reverses the worst order; it separates confused pairs exactly as far as the worst order does but swaps which member of each pair holds the larger allocation, which isolates the directional part of the claim from the separation itself.

In the worst order, colon adenocarcinoma sits at rank 2 and rectum adenocarcinoma at rank 28; lung squamous cell carcinoma sits at rank 0, with head and neck squamous cell carcinoma at rank 19 and cervical squamous cell carcinoma at rank 29; the lower-grade glioma sits at rank 5 and glioblastoma at rank 24. The mild order places each of these pairs at adjacent or near-adjacent ranks. Across the worst order, rank and balanced-arm recall correlate at a Spearman coefficient of only 0.08, so the order is not a disguised sort by difficulty, which was the failure mode of the worst-assignment study.

Each of the three orders was fitted at $\rho = 100$ on all 30 split-draw combinations, 90 fits in total, with the same patients and patch orderings as every other arm. The r1, r10, r100 and native-distribution arms of the prevalence curve and the easy-tail and hard-tail arms of the tail-assignment study are reused unchanged (Queisler, 2026b,d).

3 Evaluation

The *damage* D of an arm is its loss of test patient-macro balanced accuracy against r1, in points. Pooling and uncertainty follow the prevalence curve: equal weight per split, Bayesian bootstrap weights on test patients (Rubin, 1981) combined with resampling of draws within splits, and 95 percent percentile intervals of every paired contrast from 2,000 replicates.

3.1 Gate and calibration line

Before any of the 90 fits, the repaired score was checked with the same gate that the worst-assignment study fixed (Queisler, 2026e). G1 requires $S(\text{easy}) > S(\text{hard})$; G2 requires the mean of S over the 30 random orders of the prevalence curve to exceed $S(\text{easy})$; G3, descriptive, reports the Spearman correlation between S and single-fit damage across those random orders. In addition, a straight line of D on S was fitted to the 30 random orders alone and used to predict the damage of the easy-tail and hard-tail orders, which the line does not see, and of the three derived orders. These predictions were recorded before submission and are not refitted afterwards.

Single-fit damages are noisy. Across the 30 random orders the damage has a standard deviation of 0.65 points, and fixed orders refitted on all 30 split-draw combinations vary about as much, by 0.48 to 0.73 points per fit. Most of the per-fit variance therefore comes from the patients and test sets of each draw rather than from the order, so even a perfect score could reach only a moderate Spearman coefficient against single random fits.

3.2 Pre-registered endpoints

Three contrasts were fixed before submission, each judged by whether its 95 percent interval excludes zero and against the one-point practical threshold of the series:

- **H1** (primary): $D_{\text{worst}} - D_{\text{r100}} > 0$, where D_{r100} is the random-order damage of the prevalence curve.
- **H2**: $D_{\text{r100}} - D_{\text{mild}} > 0$.
- **H3** (direction): $D_{\text{worst}} - D_{\text{flip}} > 0$. The line predicted a difference of 0.53 points, which the design might not resolve.

Two descriptive analyses complete the evaluation: the observed against the predicted damage of all five fixed orders, and the per-class recall change for the main confused pairs.

The score deserves a caveat stated before the results. It was chosen after exploratory checks on stored outputs, from about nine variants of the weight and direction terms, and its agreement with the easy-tail and hard-tail anchors is therefore in-sample in the sense that matters: those anchors informed the choice. It differs from the failed score by one change with a clear rationale, and three variants that weight pairs symmetrically or by row share produced worst and mild orders of similar predicted damage. The three derived orders were never fitted before the choice was fixed, and they are the real test.

4 Results

The gate passes, all 90 fits converged, and all three pre-registered contrasts resolve in the predicted direction. This section reports the gate first, then the damages of the derived orders, and then the per-class recall changes behind them.

4.1 The repaired score passes the gate

G1 and G2 both hold. The hard-tail order scores 350, the easy-tail order 453, and the 30 random orders 604 on average, with a range of 475 to 771; the order of the three scores matches the observed damages, 1.56, 2.35 and 2.76 points. G3 is positive: across the random orders, score and single-fit damage correlate at a Spearman coefficient of 0.49 ($p = 0.005$, $n = 30$), and at 0.40 when each order is predicted from a line fitted without it. Given the single-fit noise of Section 3.1, a moderate correlation is what a useful score can be expected to reach.

The calibration line fitted to the random orders, $D = 0.40 + 0.00390\,S$, predicts 2.17 points for the easy-tail order and 1.77 points for the hard-tail order. Both lie inside the observed intervals, 1.88 to 2.80 and 1.20 to 1.93 points. For the three derived orders it predicts 3.91 points (worst), 3.38 points (flip) and 1.20 points (mild). The worst and mild orders score 898 and 204, which lie 4.1 and 5.5 standard deviations of the random-order scores above and below their mean, so both predictions extrapolate well beyond the range on which the line was fitted.

4.2 Derived orders span 1.3 to 4.2 points of damage

Table 1 lists the damage of every fixed order together with the random-order average. The worst order costs 4.17 points, more than any other fixed order and more than all but one of the 30 single random fits, whose damages range from 1.40 to 4.41 points. The mild order costs 1.28 points, the lowest of the fixed orders, and close to the 1.17 points of the native-distribution arm of the prevalence curve. The ranking worst > flip > random > easy > hard > mild holds in two of the three splits; in the third, the easy-tail order costs slightly more than the random orders. Table 2 gives the pre-registered contrasts. **H1** holds: the worst order costs 1.41 points more than a random order, with an interval of 0.94 to 1.88 points that excludes zero; the point estimate exceeds the one-point threshold, but the lower end of the interval falls just below it. **H2** holds: the mild order costs 1.48 points less than a random order, with an interval of 1.04 to 1.92 points that lies entirely above the threshold. The two effects are nearly symmetric around the random

Class order	S	Predicted D	Observed D (pp)		Source
Worst (argmax S)	898	3.91	4.17	(3.55 to 4.76)	this report
Flip (reversed worst)	763	3.38	3.07	(2.61 to 3.54)	this report
Random, mean over 30 orders	604	2.76	2.76	(2.34 to 3.20)	prevalence curve
Easy-tail	453	2.17	2.35	(1.88 to 2.80)	tail assignment
Hard-tail	350	1.77	1.56	(1.20 to 1.93)	tail assignment
Mild (argmin S)	204	1.20	1.28	(0.94 to 1.61)	this report

Table 1: The score orders all six assignments by damage, and a line fitted only on random orders predicts every fixed order within 0.30 points. S is Equation (1) at $\rho = 100$. Predicted D comes from the calibration line fitted to the 30 random orders before the derived orders were fitted; the random-order row is fitted, not predicted. Observed D is the loss of test patient-macro balanced accuracy against r1, pooled over 30 fits, with 95 percent bootstrap intervals. Random and sorted orders are reused from Queisler (2026b) and Queisler (2026d).

average, and the full range between the worst and the mild order is 2.89 points, larger than the 2.76 points the ratio costs on average.

H3 holds as well: the worst order costs 1.09 points more than its reversal, with an interval of 0.42 to 1.78 points. This contrast was expected to be the hardest to resolve, since the line predicted only 0.53 points. The flip order separates each confused pair by exactly the same number of ranks as the worst order, so the difference is due to which member of each pair receives the larger allocation. At the same time, the flip order lies 0.32 points above the random average, a difference without a pre-registered interval but consistent with separation alone, regardless of direction, carrying part of the effect.

The descriptive comparison of the mild and hard-tail orders stays unresolved, at -0.29 points (-0.61 to 0.03), as expected from a predicted difference of -0.57 points and a paired interval half-width of about 0.32 points.

Endpoint	Contrast	Estimate (pp)	95% interval
H1	$D_{\text{worst}} - D_{\text{r100}}$	1.41	0.94 to 1.88
H2	$D_{\text{r100}} - D_{\text{mild}}$	1.48	1.04 to 1.92
H3	$D_{\text{worst}} - D_{\text{flip}}$	1.09	0.42 to 1.78
Descriptive	$D_{\text{mild}} - D_{\text{hard}}$	-0.29	-0.61 to 0.03

Table 2: All three pre-registered contrasts exclude zero; H2 exceeds the one-point threshold with its whole interval, and H1 with all but the lower end of its interval. Paired contrasts of balanced-accuracy damage at $\rho = 100$, pooled over 30 fits per arm with 95 percent bootstrap intervals.

Figure 2 places the fitted orders against the random cloud. The derived orders lie on the extension of the random-order trend at both ends. Regressing the observed on the predicted damage of the five fixed orders gives a slope of 1.02, and the mean absolute prediction error is 0.21 points both over all five and over the three derived orders alone. Although the extrapolation caveat anticipated that the line might overshoot, it does not at either extreme: it underestimates the worst order by 0.26 points and the mild order by 0.08 points, and its largest miss is the flip order, which it overestimates by 0.30 points. Every prediction for the three derived orders lies inside the observed interval of that order.

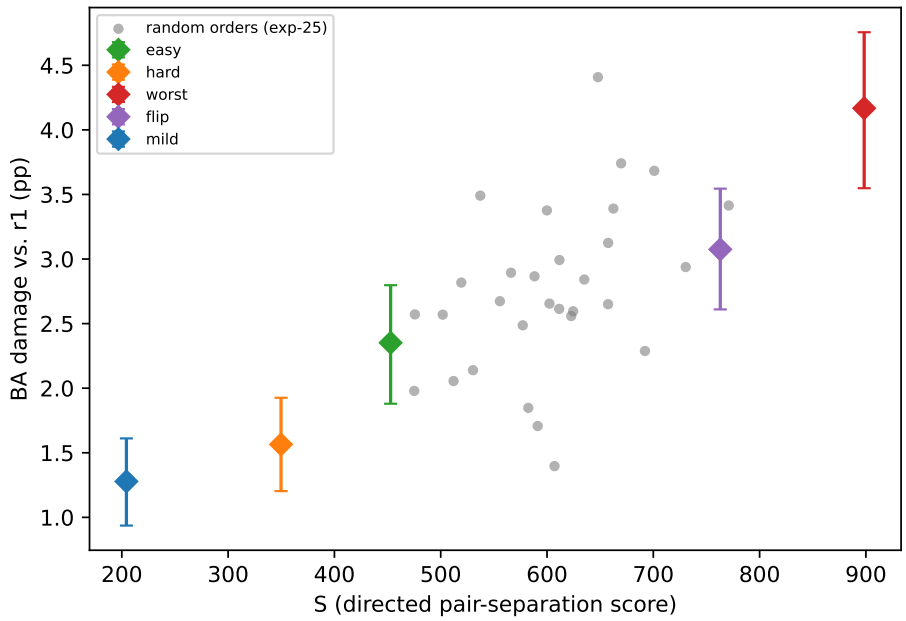


Figure 2: The derived orders extend the random-order trend in both directions. Observed balanced-accuracy damage against the rate-weighted score at $\rho = 100$. Grey points: the 30 random class orders of the prevalence curve, one per fit. Diamonds with 95 percent bootstrap intervals, each pooled over 30 fits: hard-tail (orange) and easy-tail (green) orders of the tail-assignment study, and the mild (blue), flip (purple) and worst (red) orders fitted here.

4.3 Recall moves with allocation relative to the confusion partner

The damage of the worst order is concentrated. Nine classes lose more than ten points of recall under it, against two under the mild order, and the largest single loss is 44.2 points against 13.2 points. Table 3 follows the main confusion groups through the three derived orders. Three groups are mutual pairs, in which each class is the other's main confusion partner. The fourth is a cluster around lung squamous cell carcinoma (LUSC): head and neck squamous cell carcinoma (HNSC), cervical squamous cell carcinoma (CESC) and lung adenocarcinoma (LUAD) are each most often predicted as LUSC, so all three are paired with the same class and appear in one block.

Within every group, the member placed far below its partner loses recall and the member placed above it gains, and the roles exchange when the order is reversed. Rectum adenocarcinoma loses 44.2 points at rank 28 below colon adenocarcinoma and gains 22.7 points at rank 1 above it. In the mild order both sit in the tail, at ranks 28 and 29, and lose 11.1 and 3.7 points, far less than either loses when separated from the other. The squamous cluster and the gliomas behave in the same way.

The direction effect of H3 is visible in the asymmetry within pairs. Adrenocortical carcinoma loses 24.5 points when it sits at rank 27 below hepatocellular carcinoma, whereas hepatocellular carcinoma loses only 5.9 points in the reversed position. The two classes are each other's main confusion partner, but under balanced training adrenocortical carcinoma is misclassified far more often, with a recall of 72.9 against 85.4 percent; the rate-weighted score encodes this asymmetry through $w_{c \to d}$. Figure 3 generalizes the table to all 30 classes: under the mild order almost every class sits within a few ranks of its main confusion partner and its recall change stays within 13 points, whereas under the worst order half of the classes sit 15 or more ranks from their partner and recall changes spread from -44 to $+29$ points.

			Worst		Flip		Mild	
Class	Partner	h_c (%)	Rank	Δ	Rank	Δ	Rank	Δ
COAD	READ	51.6	2	+28.8	27	−34.2	28	−11.1
READ	COAD	60.4	28	−44.2	1	+22.7	29	−3.7
LUSC	HNSC	54.6	0	+22.2	29	−38.2	23	−9.2
HNSC	LUSC	69.0	19	−14.7	10	+1.9	24	−5.3
CESC	LUSC	46.7	29	−27.4	0	+18.0	22	−0.9
LUAD	LUSC	65.3	26	−20.1	3	+11.5	16	+2.4
LGG	GBM	78.7	5	+9.3	24	−12.5	9	+1.0
GBM	LGG	77.1	24	−17.1	5	+9.3	10	+1.3
LIHC	ACC	85.4	8	+2.3	21	−5.9	5	+1.9
ACC	LIHC	72.9	27	−24.5	2	+7.3	6	+3.2

Table 3: Classes placed far below their main confusion partner lose large amounts of recall, and the loss is not symmetric within a pair. Partner is the class that the balanced arm most often predicts for the row's class; h_c is balanced-arm recall; rank 0 holds 2,843 patches and rank 29 holds 28; Δ is the pooled change in own recall against r1, in points. Blocks group classes by partner. Three blocks are mutual pairs: colon and rectum adenocarcinoma, lower-grade glioma (LGG) and glioblastoma (GBM), and liver hepatocellular (LIHC) and adrenocortical carcinoma (ACC). The second block is a cluster rather than a pair: HNSC, CESC and LUAD all have LUSC as their partner, and LUSC has HNSC.

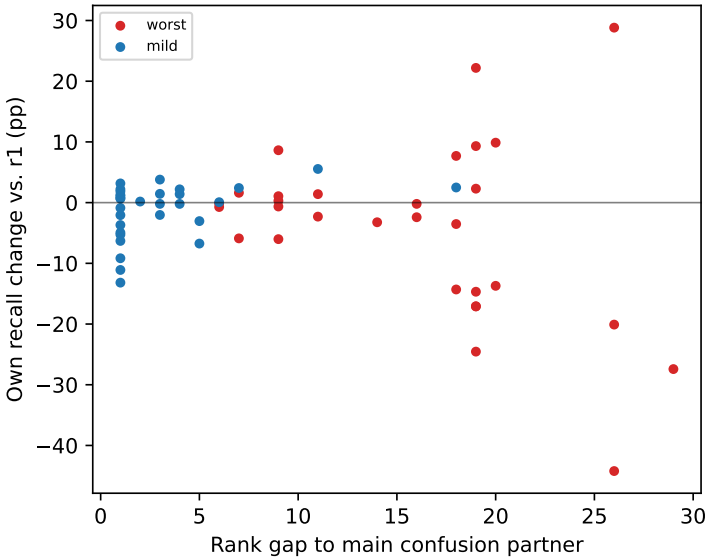


Figure 3: Recall changes spread widely only when a class is far in rank from its main confusion partner. Each point is one class under the worst (red) or mild (blue) order: the rank gap to the class it is most often predicted as in the balanced arm, against the pooled change in its own recall against r1.

5 Discussion

The mechanism that the tail-assignment study read off two fitted orders (Queisler, 2026d) now predicts damage. A single scalar over class orders, fitted on random orders alone, ranks six assignments correctly and predicts the damage of three assignments it was never fitted on within 0.30 points. The orders it produces move the damage by 1.4 and 1.5 points in the two directions around the random average; both shifts exceed the one-point threshold of the series, the mild shift with its whole interval and the worst shift with all but the lower end of its interval. The pairwise account is therefore not only a description of which classes moved in one experiment. It is a quantitative predictor of how much a whole assignment costs.

Two parts of that account are separately supported. Separation of confused pairs matters: the flip order separates them as far as the worst order and is still more damaging than a random order. Direction matters as well: which member of a pair holds the larger allocation accounts for another 1.09 points. The per-class results explain both effects. A class that is often predicted as a well-supported partner gives up recall to it, and the size of the loss depends on how often the balanced classifier already makes that error and how much recall the class has. The class at the tail does not need to be difficult on its own; rectum adenocarcinoma and adrenocortical carcinoma lose most of their recall only when their specific partner is abundant.

The contrast with the failed score is instructive. The two scores differ only in whether a class's confusion weights sum to one or to its error rate. With row-normalized weights, a class the classifier almost never confuses still contributed a full unit of weight, and the score responded to the monotone alignment of recall and rank that sorting creates (Queisler, 2026e). With rate weights, such a class contributes almost nothing, and the score responds to where confusion actually occurs. The worst order it produces is essentially unrelated to difficulty, with a correlation of 0.08 between rank and balanced recall.

For the questions of this series, the result reframes the damage of a long-tailed allocation. At $\rho = 100$ and a fixed budget, the random-order average of 2.76 points is one point on a range of at least 1.28 to 4.17 points spanned by the assignment alone. Which classes are rare is thus as consequential as how rare they are. The practical corollary is that a long-tailed data set can be more or less harmful depending on whether confused classes are rare together or apart, and that the balanced classifier's confusion rates indicate in advance which case applies. Even the mildest assignment found still costs 1.28 points, so part of the cost of the ratio remains whichever classes are made rare.

6 Limitations

The score was chosen after exploratory checks on stored outputs, among about nine variants, and the easy-tail and hard-tail anchors informed that choice. Its agreement with those two anchors is therefore not an independent test. The three derived orders are, since they were fixed and their predictions recorded before any of them was fitted; but they are three orders, derived from the same score, and they cannot establish how well the score ranks the space of assignments between them.

The worst and mild orders are local extremes found by hill climbing, not proven optima. Some assignment may be more damaging than the worst order reported here, so 4.17 points is a lower bound on the worst case rather than an estimate of it. The score inputs, confusion rates and balanced recall, are taken from balanced-arm predictions on the same test patients on which damage is evaluated. This is appropriate for a study of mechanism, but a practitioner would have to estimate confusion from validation data, and the predictive accuracy reported here is an upper bound for that setting.

All derived orders were fitted at $\rho = 100$. The score's ordering of assignments does not depend on the ratio, but whether the damage ordering carries over to milder ratios is not tested,

and the single-fit noise of the random orders caps the within-family correlation that any score can reach. All results are conditional on twenty patients per class, frozen Virchow2 features, multinomial logistic regression, and the three overlapping splits of TCGA-UT; on data sets whose classes collapse in the tail, such as BRACS, headroom rather than confusion structure dominated (Queisler, 2026c), and the score has not been tested there.

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